



PROJECT REPORT

HANDS FREE: GESTURE CONTROLLED VIRTUAL MOUSE SYSTEM

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INTRODUCTION

This project is a computer vision-based application that allows users to control their computer using hand gestures instead of a physical mouse. It uses Python along with OpenCV, MediaPipe, and PyAutoGUI to detect hand movements through a webcam and translate them into cursor actions.

The aim of this project is to provide a touch-free and intuitive way of interacting with a computer. It demonstrates how real-time hand tracking and gesture recognition can be used to replace traditional input devices, making human-computer interaction more natural and accessible.

Problem Statement

Traditional computer interaction relies heavily on physical input devices such as a mouse or touchpad. While effective, these methods require direct contact and may not always be convenient or accessible. In situations such as presentations, accessibility for users with limited mobility, or environments where touch-free interaction is preferred, existing solutions are either expensive or overly complex.

This project addresses the problem of enabling users to control a computer system without physical contact, using only natural hand gestures captured through a webcam.

Choice of Problem Statement

The importance of this problem lies in the growing need for more intuitive and accessible human-computer interaction systems. As technology advances, there is a shift toward natural interfaces such as gesture, voice, and vision-based controls.

A gesture-controlled system has practical applications in:

- Accessibility for individuals with physical limitations
- Touchless environments where hygiene is important
- Interactive systems such as smart displays and presentations
- Emerging technologies like augmented and virtual reality

By developing a low-cost, software-based solution using widely available tools, this project demonstrates how advanced interaction systems can be made more accessible.

Approach to the Solution

The solution is built as a real-time computer vision system that translates hand gestures into mouse actions. The system follows a structured pipeline:

Camera Input → Hand Detection → Gesture Recognition → System Control

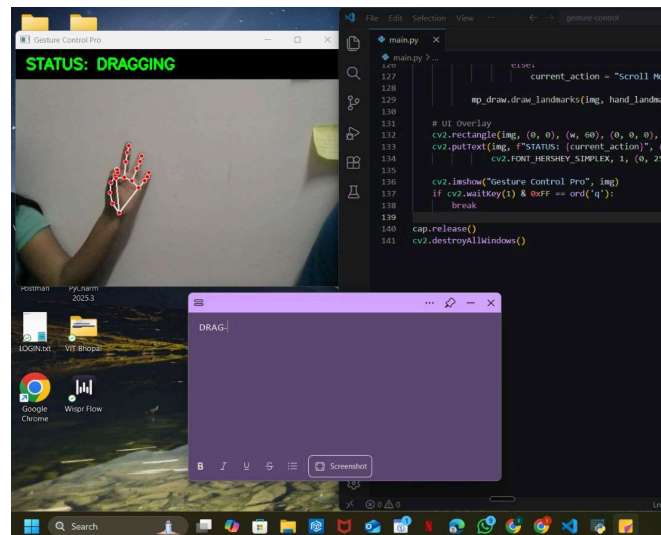
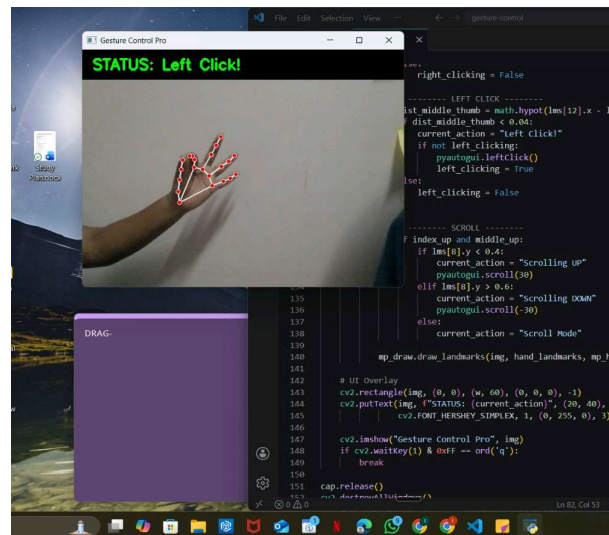
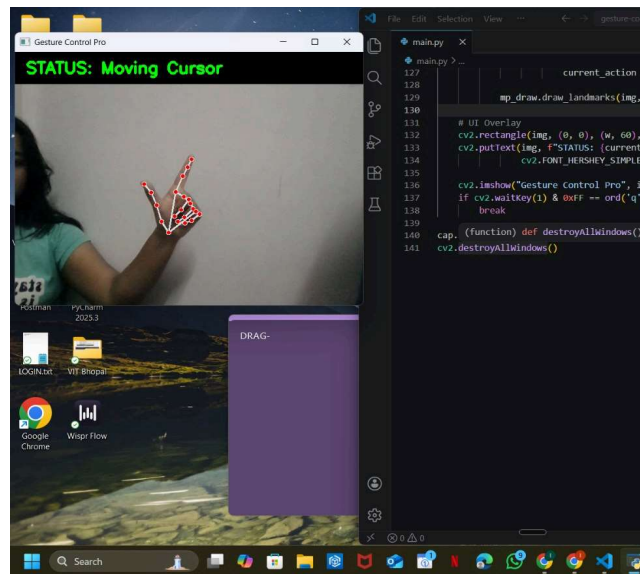
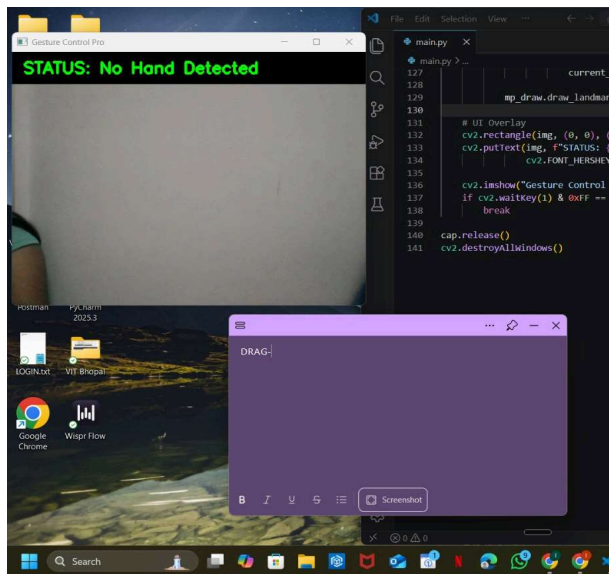
- Video input is captured using OpenCV through a webcam
- Hand tracking is performed using MediaPipe, which detects 21 key landmarks on the hand
- Finger positions and distances between landmarks are analyzed to identify gestures
- These gestures are mapped to mouse actions such as movement, clicking, scrolling, and dragging using PyAutoGUI

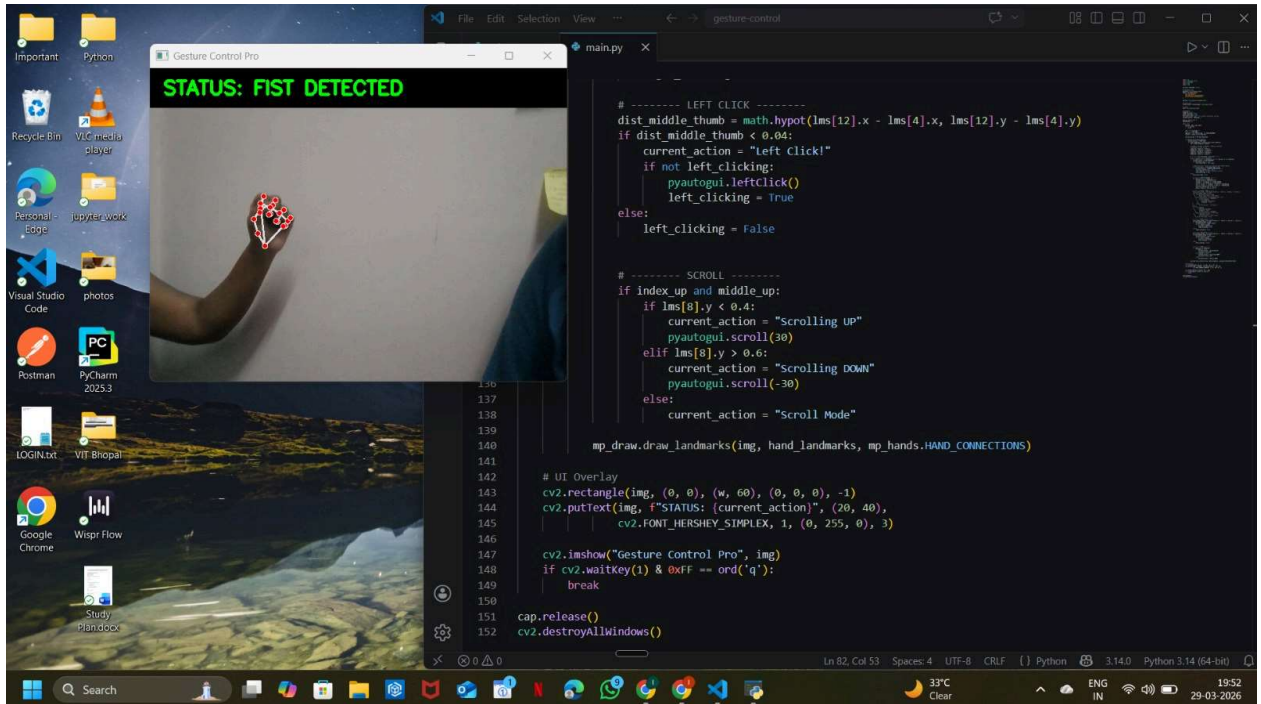
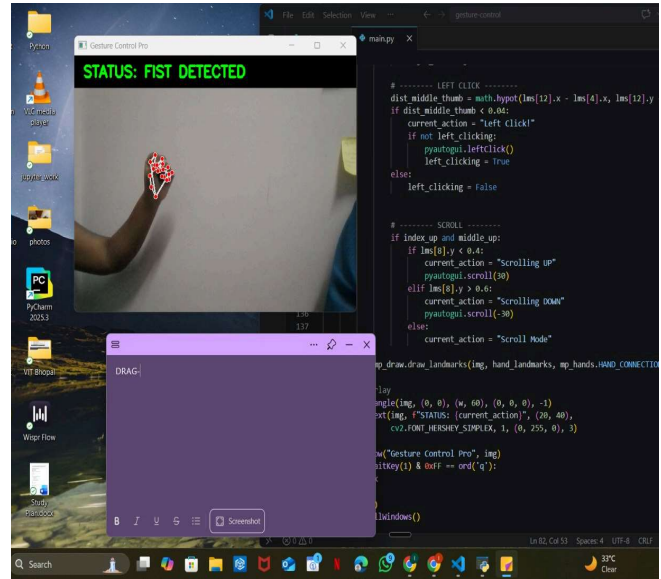
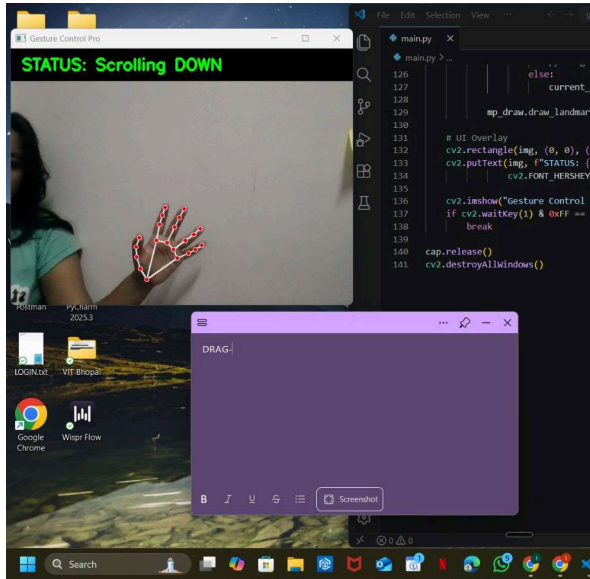
Different gestures are defined based on finger states:

- Index finger = cursor movement
- Index + middle finger = scrolling
- Thumb + index pinch = left click and drag
- Thumb + middle pinch = right click
- Closed fist = close active application

The system operates continuously in real-time, updating actions based on the user's hand movements.

Output Screenshots





Key Design Decisions

Several important decisions were made to balance performance, simplicity, and usability:

- Use of MediaPipe instead of a custom ML model:
This avoids the complexity of training a model while still providing accurate and fast hand tracking.
- Single-hand tracking:
Limiting detection to one hand reduces computational load and avoids ambiguity in gesture recognition.
- Distance-based gesture detection:
Gestures like clicking are detected using the distance between fingertips, which is computationally efficient and reliable.
- Smoothing of cursor movement:
A smoothing algorithm was applied to reduce jitter and create a more natural cursor experience.
- Real-time visual feedback:
A status bar was included to display current actions, improving usability and debugging.

These decisions ensured that the system remains responsive, and easy to use.

Challenges Faced

The development process involved several challenges:

- **Gesture overlap and unintended actions:**
Multiple gestures being detected simultaneously caused incorrect outputs. This was addressed by restructuring the logic to prioritize gestures.
- **Sensitivity tuning:**
The threshold for detecting pinch gestures required careful adjustment to avoid false clicks.
- **Cursor instability:**
Initial cursor movement was shaky due to rapid position changes. This was resolved using smoothing techniques.
- **Timing conflicts:**
Differentiating between single click, double click, and drag required precise timing control.
- **Code structure issues:**
Managing multiple conditions and gesture states made the logic complex and required careful debugging.

Learnings and Takeaways

This project provided valuable insights into both technical and practical aspects of system development:

- Gained hands-on experience with computer vision and real-time processing
- Learned how landmark-based tracking works using MediaPipe
- Understood how software can simulate hardware interactions through automation tools
- Improved problem-solving skills, especially in debugging and optimizing logic
- Realized the importance of user experience in system design
- Learned how small adjustments, such as smoothing and timing, can significantly impact usability

Overall, the project demonstrated how multiple technologies can be integrated to create an innovative and functional system.

Future Enhancements

- Implement multi-hand gesture recognition
- Introduce gesture-based mode switching to reduce conflicts
- Improve accuracy using machine learning-based gesture classification
- Add a graphical interface for user customization
- Optimize performance for lower-end systems
- Enhance robustness under varying lighting conditions

Conclusion

The gesture-controlled virtual mouse successfully demonstrates a touch-free method of interacting with a computer using only hand gestures. The system is efficient, low-cost, and practical, showcasing the potential of computer vision in everyday applications.

While there are areas for improvement, the project achieves its goal of creating an intuitive and functional alternative to traditional input devices.